

Statistics

Lecture 04

Simultaneous random variables

Simultaneous (pairwise) random variables (X,Y)

Discrete (X,Y)

- simultaneous pmf
- marginal pmf's
- conditional pmf

Continuous (X,Y)

- simultaneous pdf
- marginal pdf's
- conditional pdf

Special case: independent (X,Y)

Generalization to more than two simultaneous random variables

Covariance and correlation between X and Y

Short introduction to example of 2D random variable: bivariate normal distribution

Linear combinations of random variables:

- general case: mean and variance
- special case: average of independent random variables

General function of random variable
